

- ❑ Topic of the session: Advanced Java Programming
- ❑ **UNIT 9: JSP (Java Server Pages)**
- ❑ **University of Kigali (UoK)**
- ❑ Trainer name: Dr. NTEZIRIZA NKERABAHIZI Josbert

Java Server Pages



UNIT 9. JSP (Java Server Pages)

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9.1 Installation

Tool used:

- **Eclipse 4.6 (NEON)**
- **Tomcat Web Server 8.x**

To begin with JSP, you need to have the knowledge of **Servlet**.

Install Tomcat Web Server

To begin with JSP, you need to download **Tomcat Web Server** and declare it with Eclipse. You can see the instructions here:

- **Install Tomcat Server for Eclipse**
(<https://o7planning.org/10209/install-tomcat-server-for-eclipse>)

Create Project to start with JSP

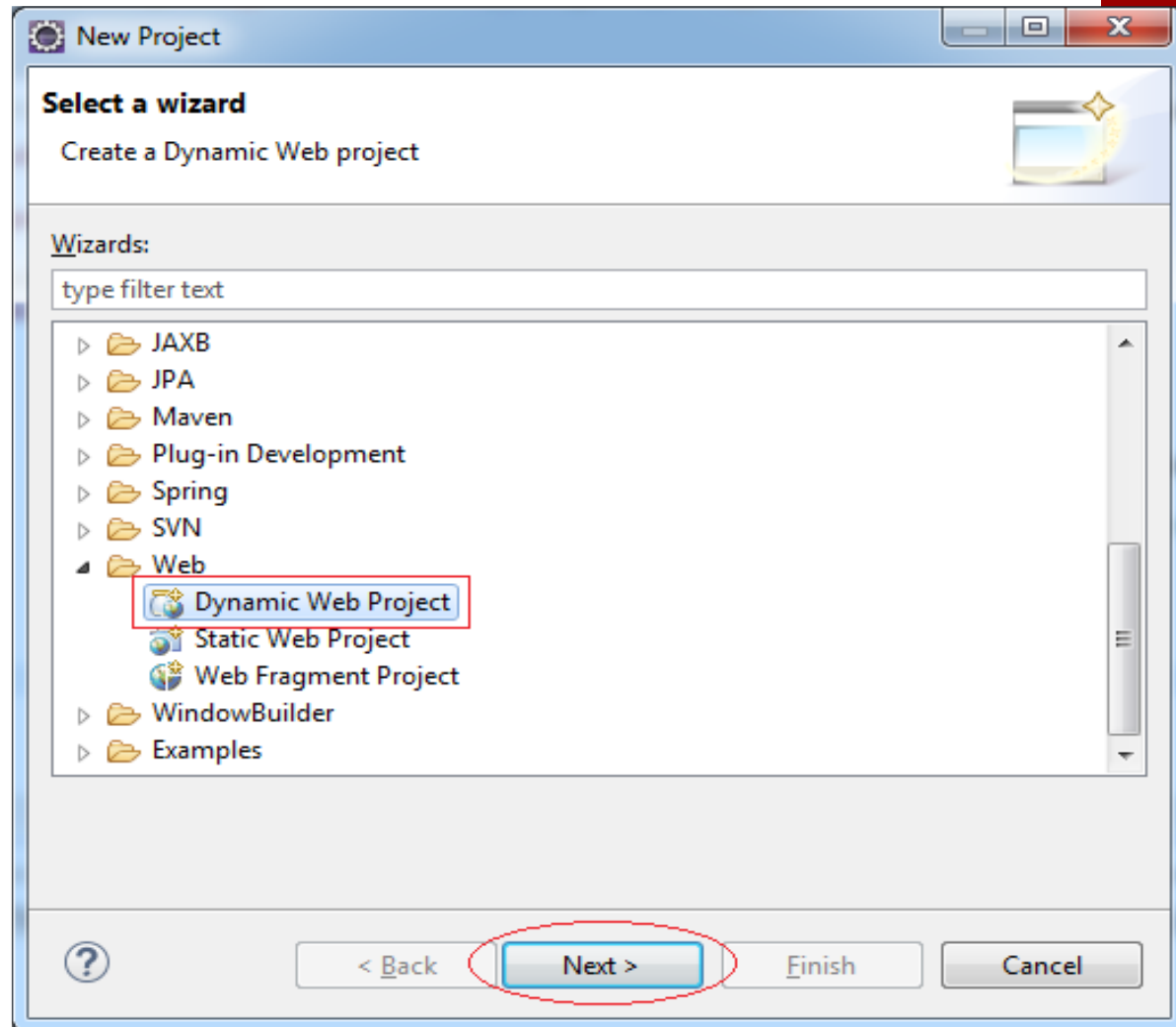
In Eclipse select:

- **File/New/Project...**

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20.1 Installation



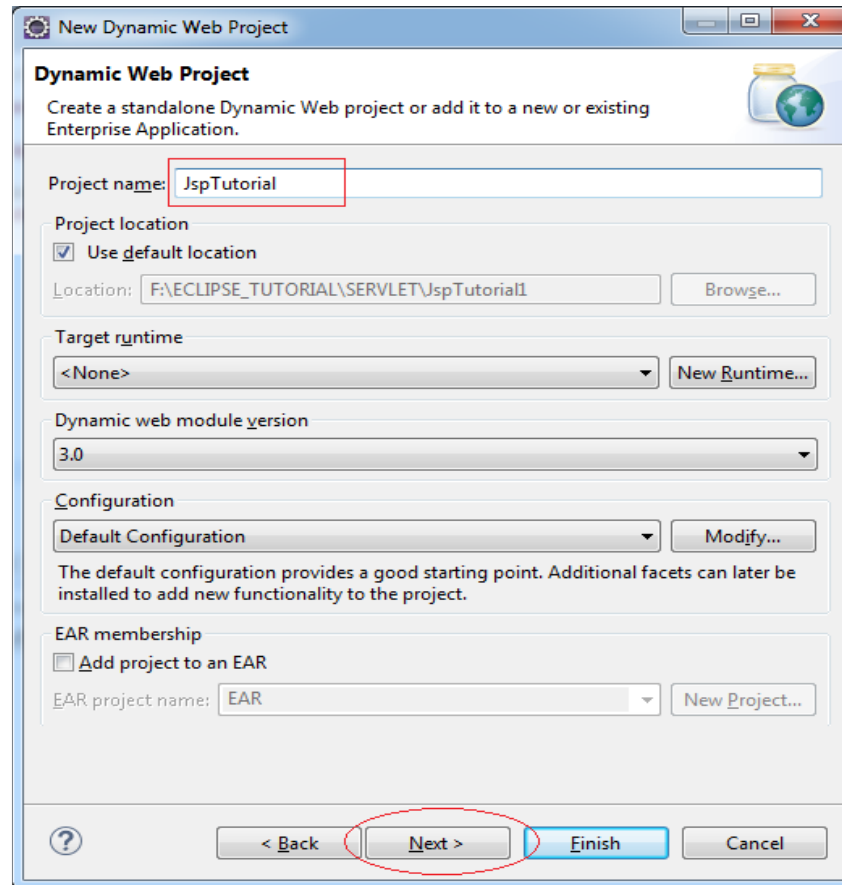
UNIT 9. JSP (Java Server Pages)

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20.1 Installation

Enter:

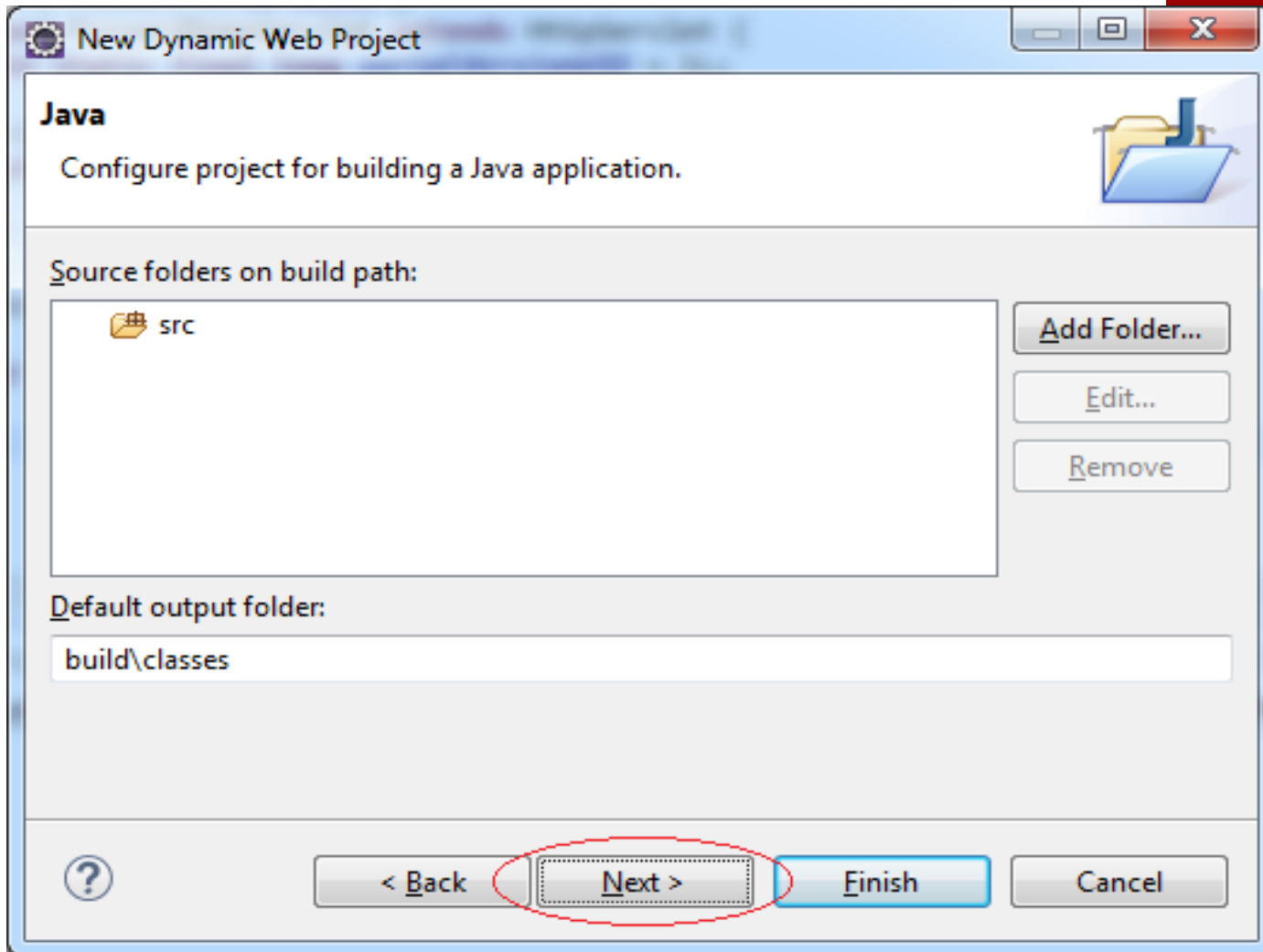
- **Project name:** JspTutorial



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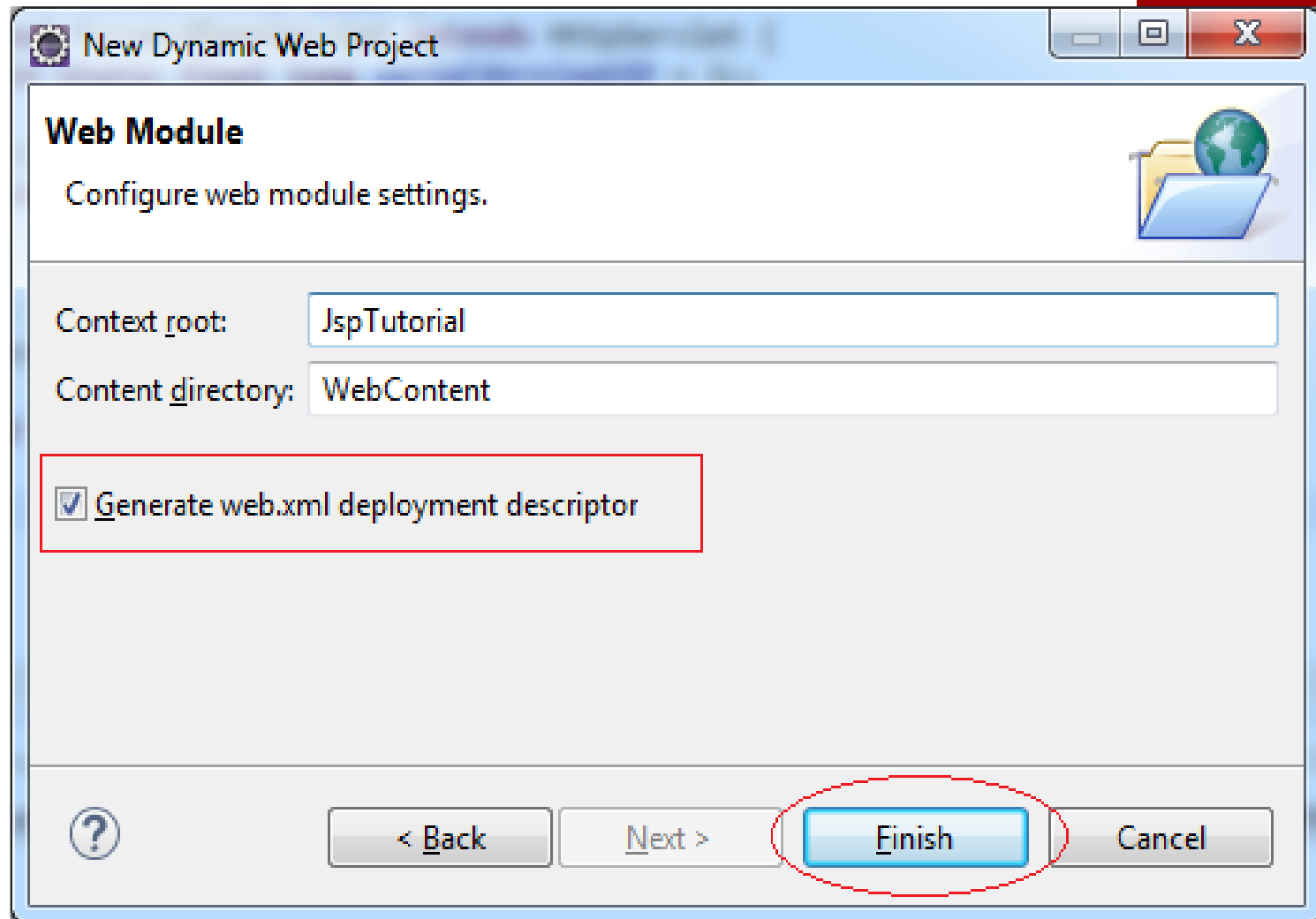
20.1 Installation



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20.1 Installation

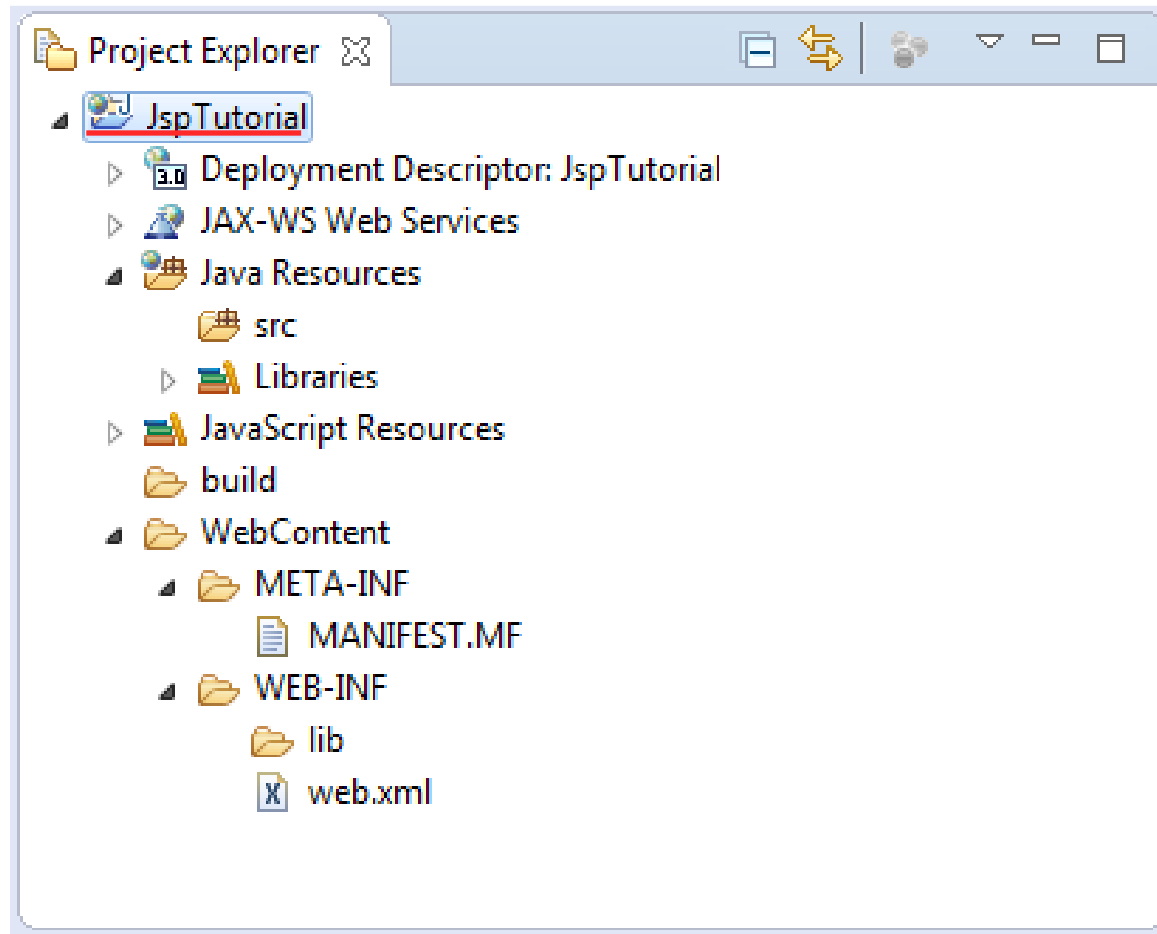


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20.1 Installation

Your Project was created

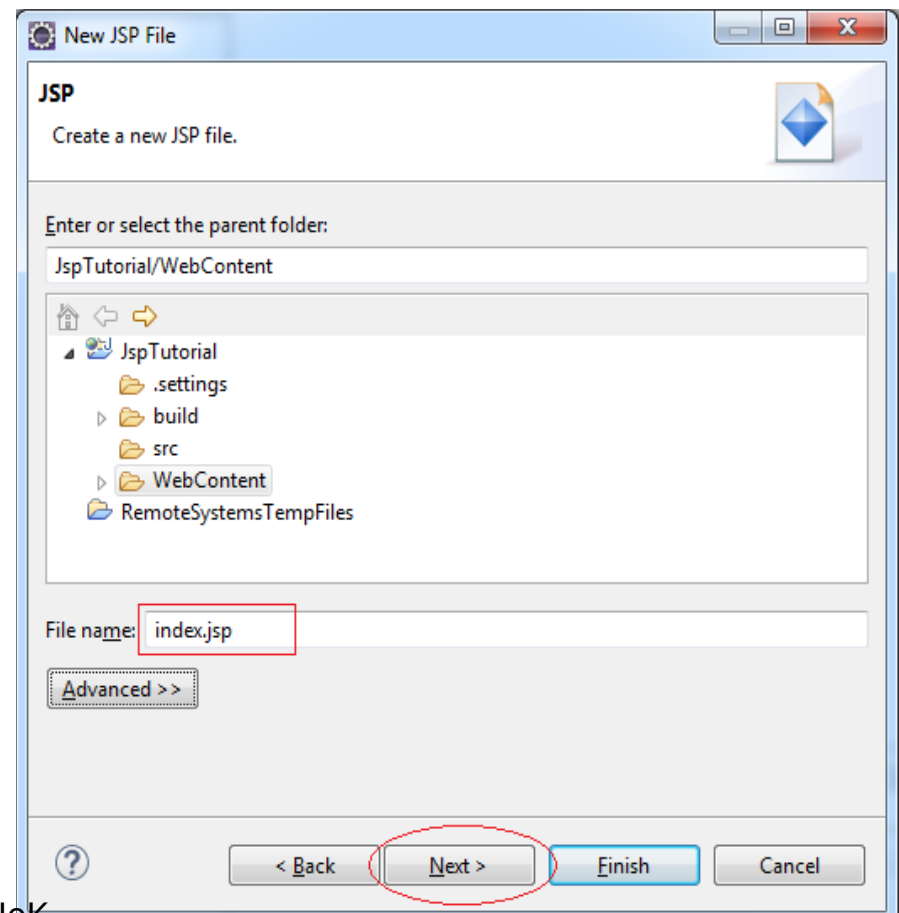
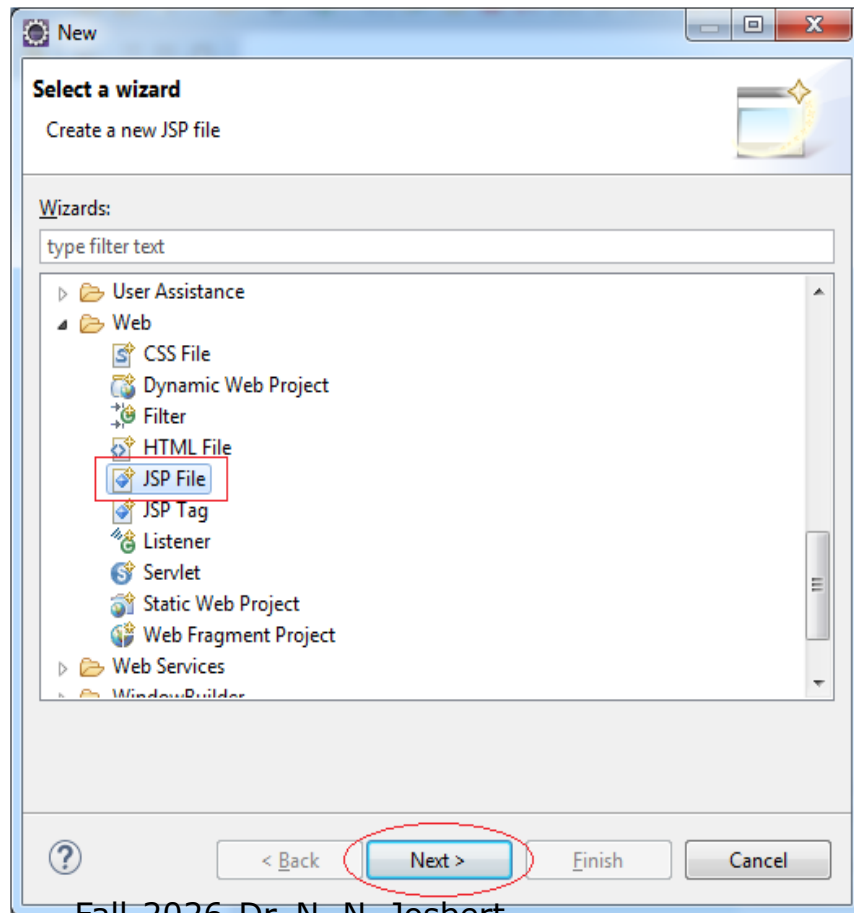


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21.1 Installation

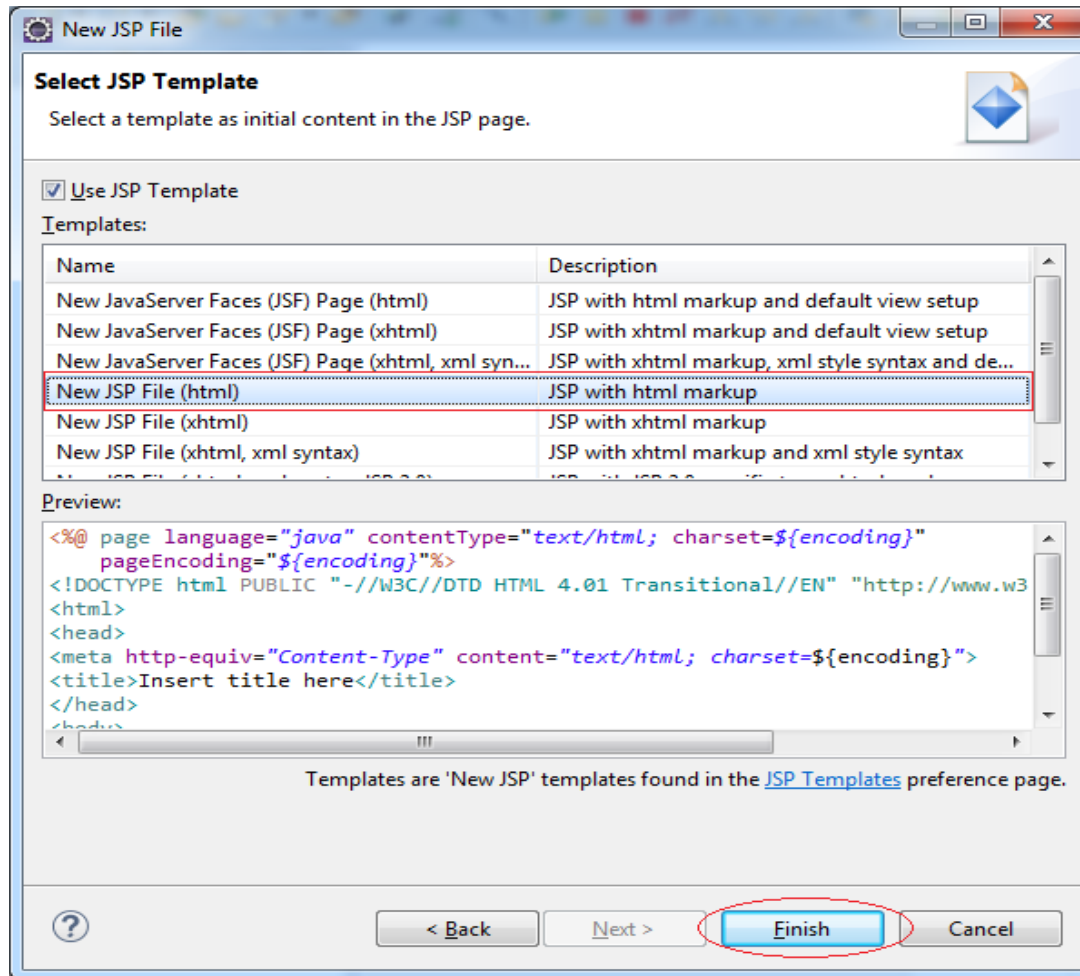
- **Create JSP file:**
 - **File/New/Other...**



UNIT 9. JSP (Java Server Pages)

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20.1 Installation

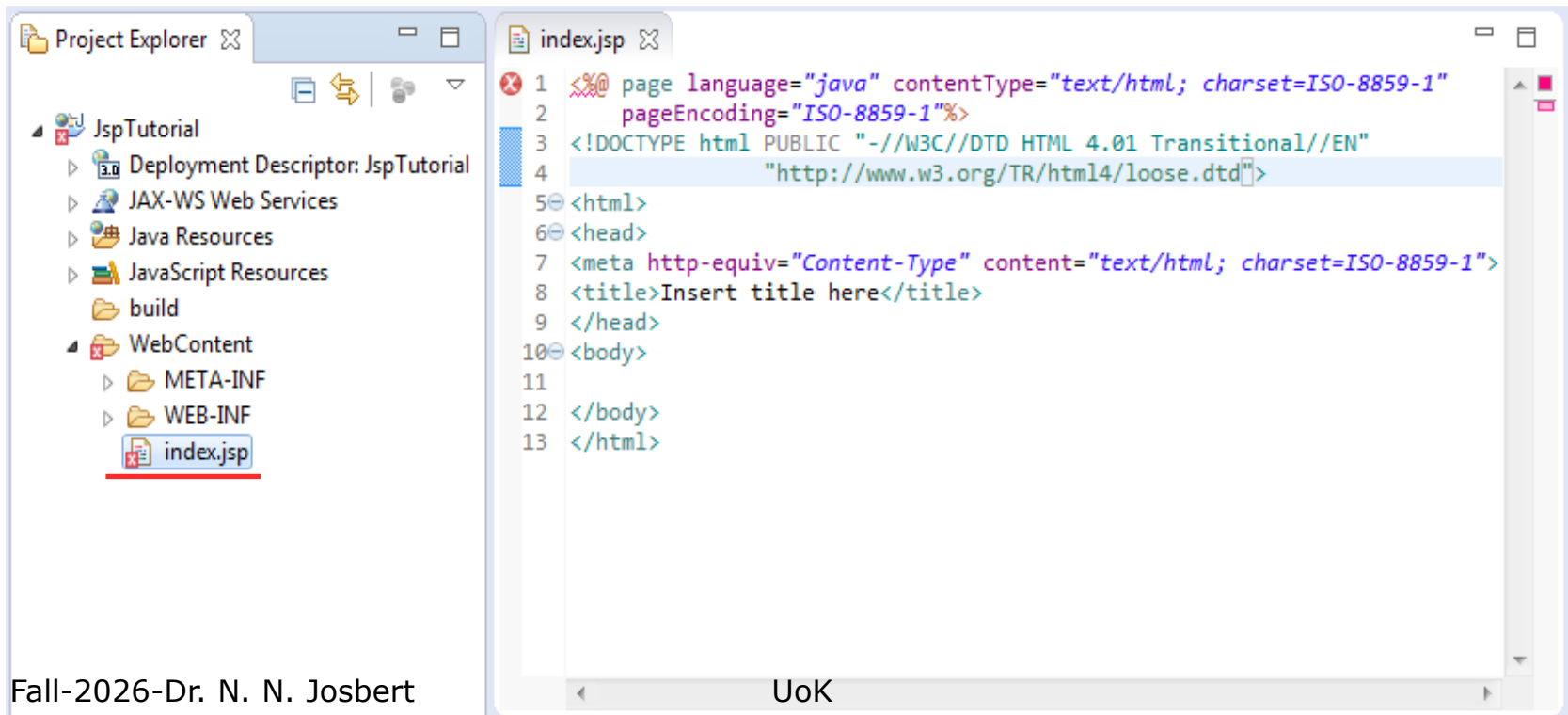


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20.1 Installation

index.jsp file is created, but there is a notification of error. Don't worry about that. It is the result of not declaring **Servlet** library. We will use the **Servlet** library of **Tomcat**, along with its configuration of **Web Server** to run this Project.



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20.1 Installation

Edit content *index.jsp* file:

index.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
```

```
pageEncoding="ISO-8859-1"%>
```

```
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
```

```
"http://www.w3.org/TR/html4/loose.dtd">
```

```
<html>
```

```
<head>
```

```
<meta http-equiv="Content-Type" content="text/html; charset=ISO-8859-1">
```

```
<title>Insert title here</title>
```

```
</head>
```

```
<body>
```

```
<h1>Hello JSP</h1>
```

```
<%
```

```
java.util.Date date = new java.util.Date();
```

```
%>
```

```
<h2>
```

Now is

```
<%=date.toString()%>
```

```
</h2>
```

```
</body>
```

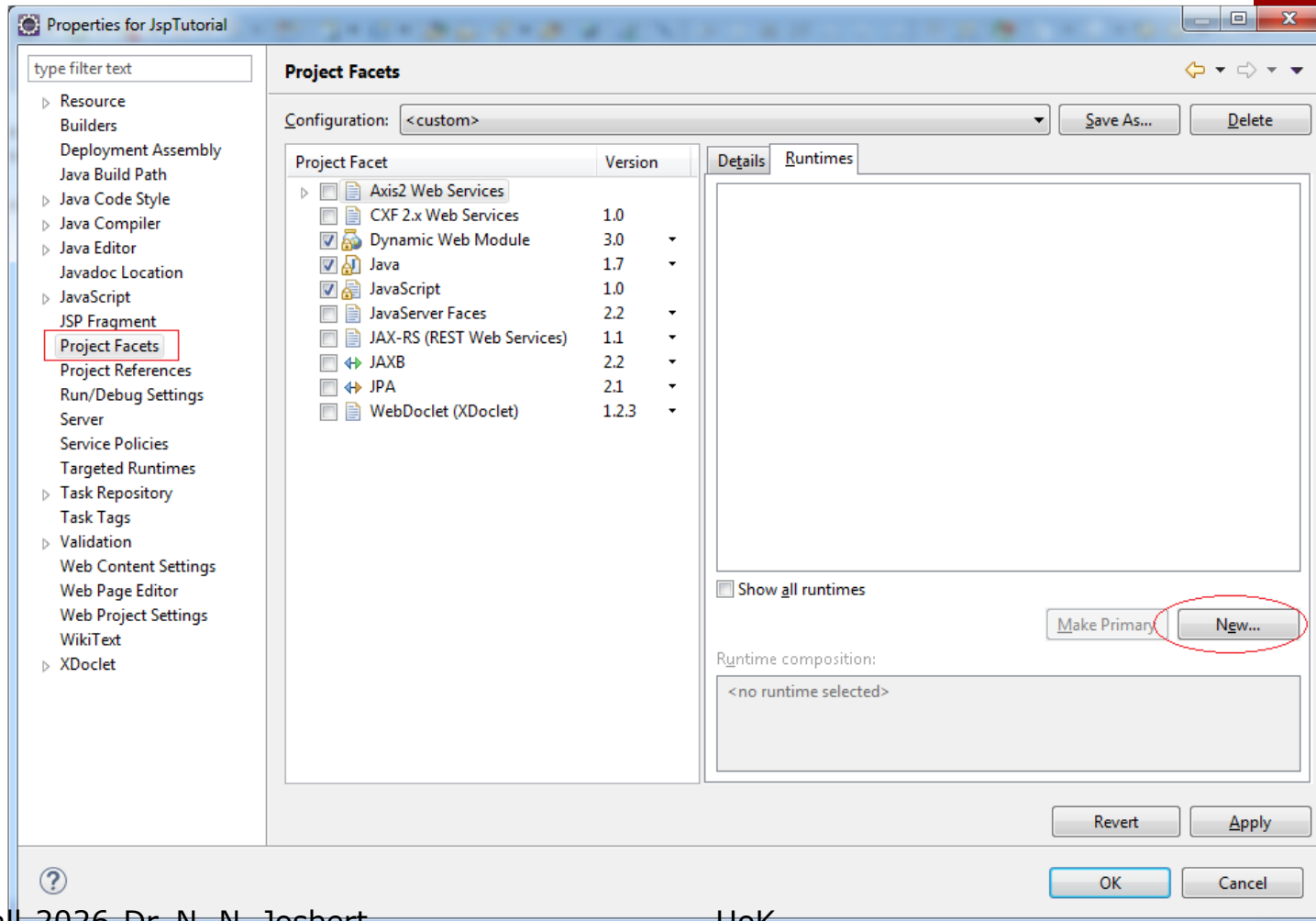
```
</html>
```

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20.2 Configuring Tomcat Web Server and run WebApp

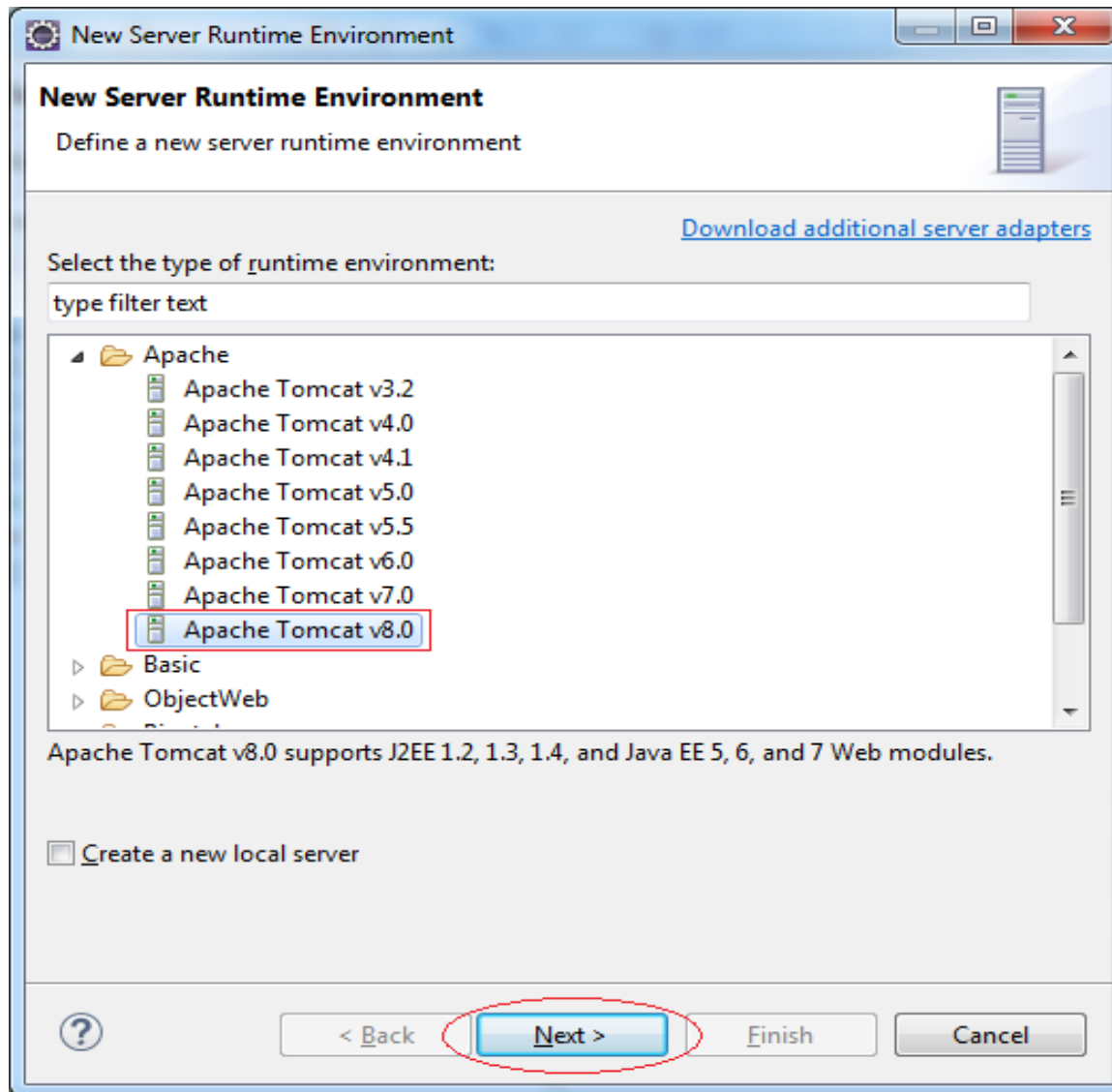
Right-click the **JspTutorial** project and select **Properties**



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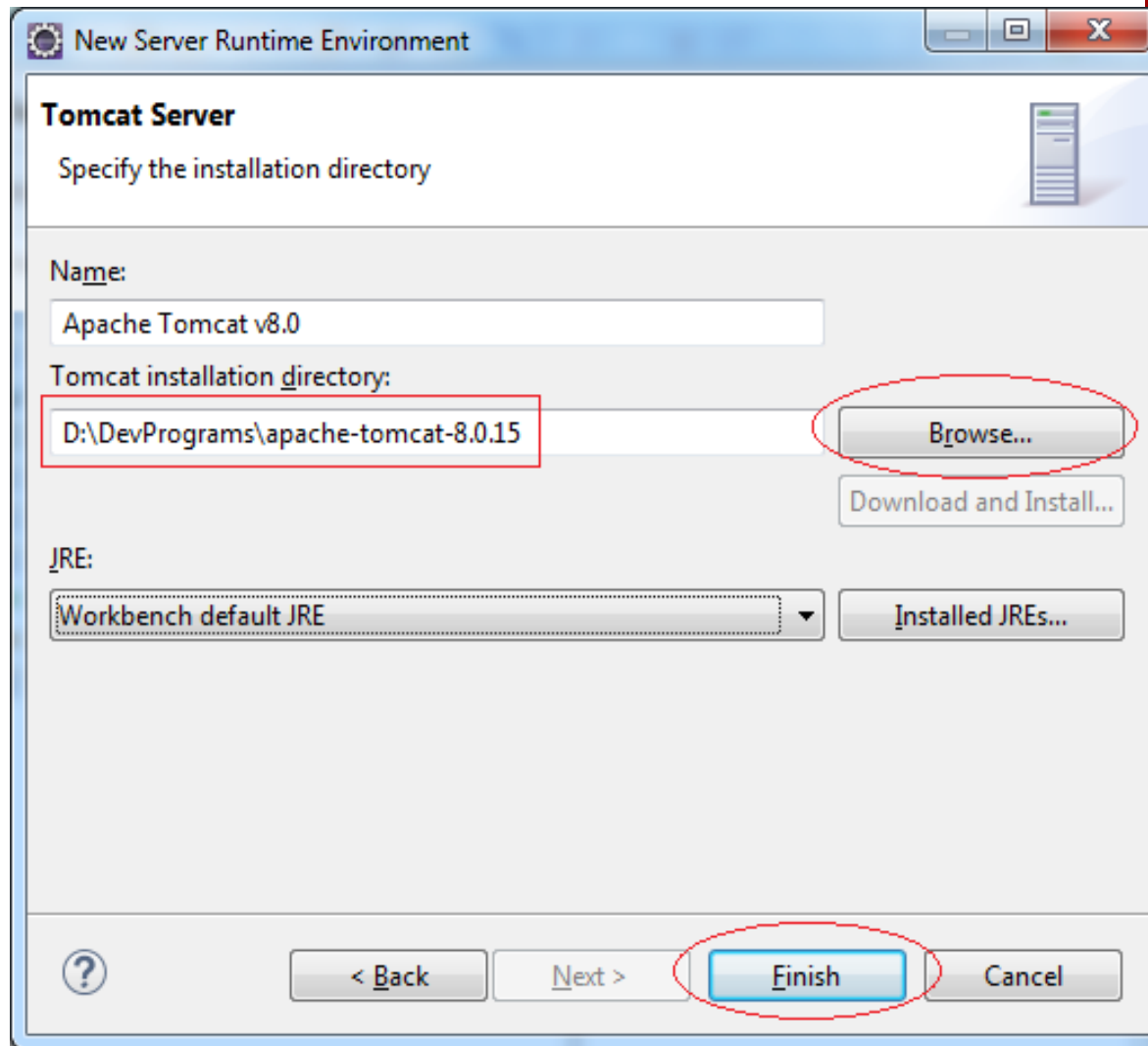
20.2 Configuring Tomcat Web Server and run WebApp



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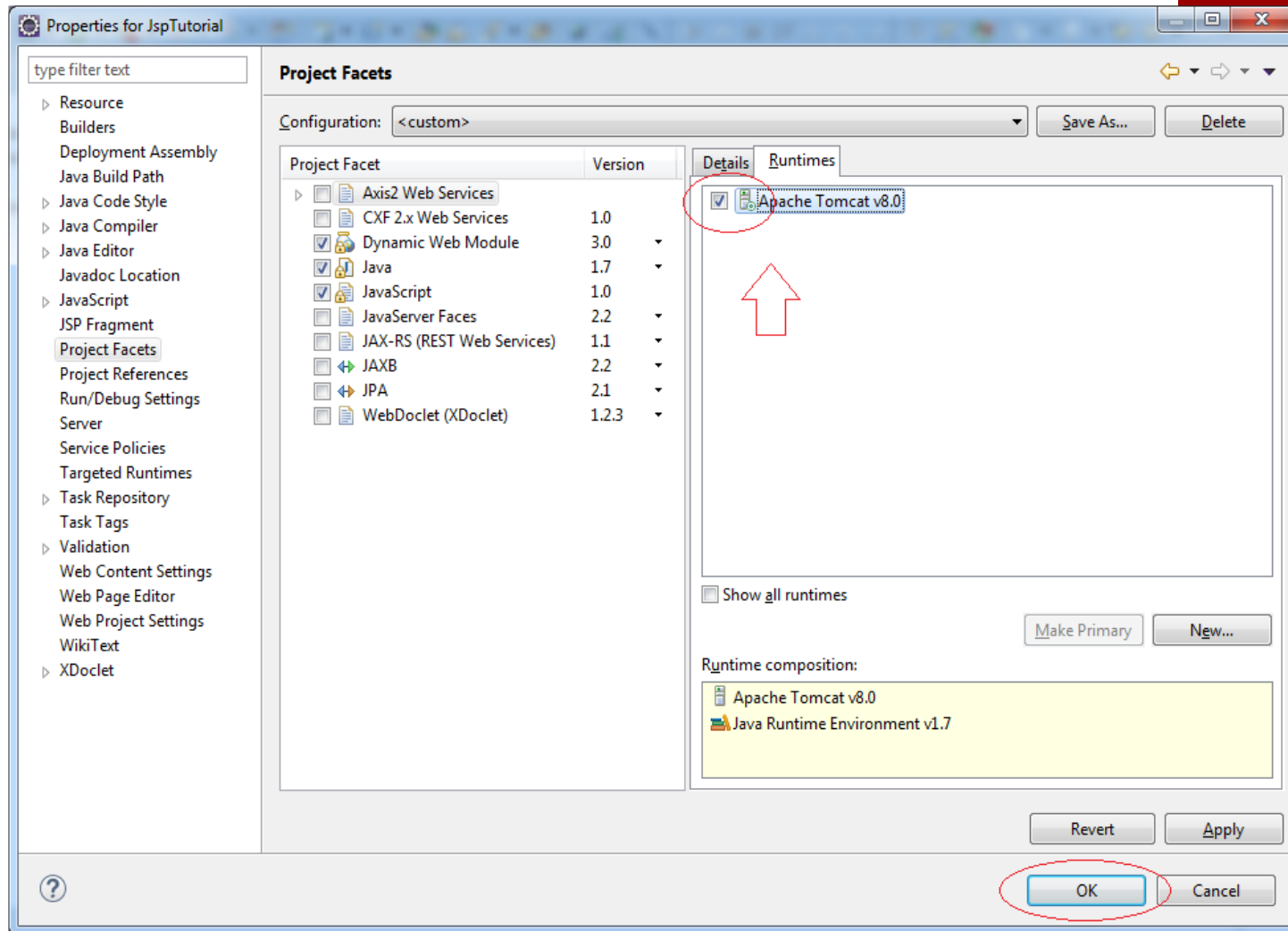
20.2 Configuring Tomcat Web Server and run WebApp



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20.2 Configuring Tomcat Web Server and run WebApp

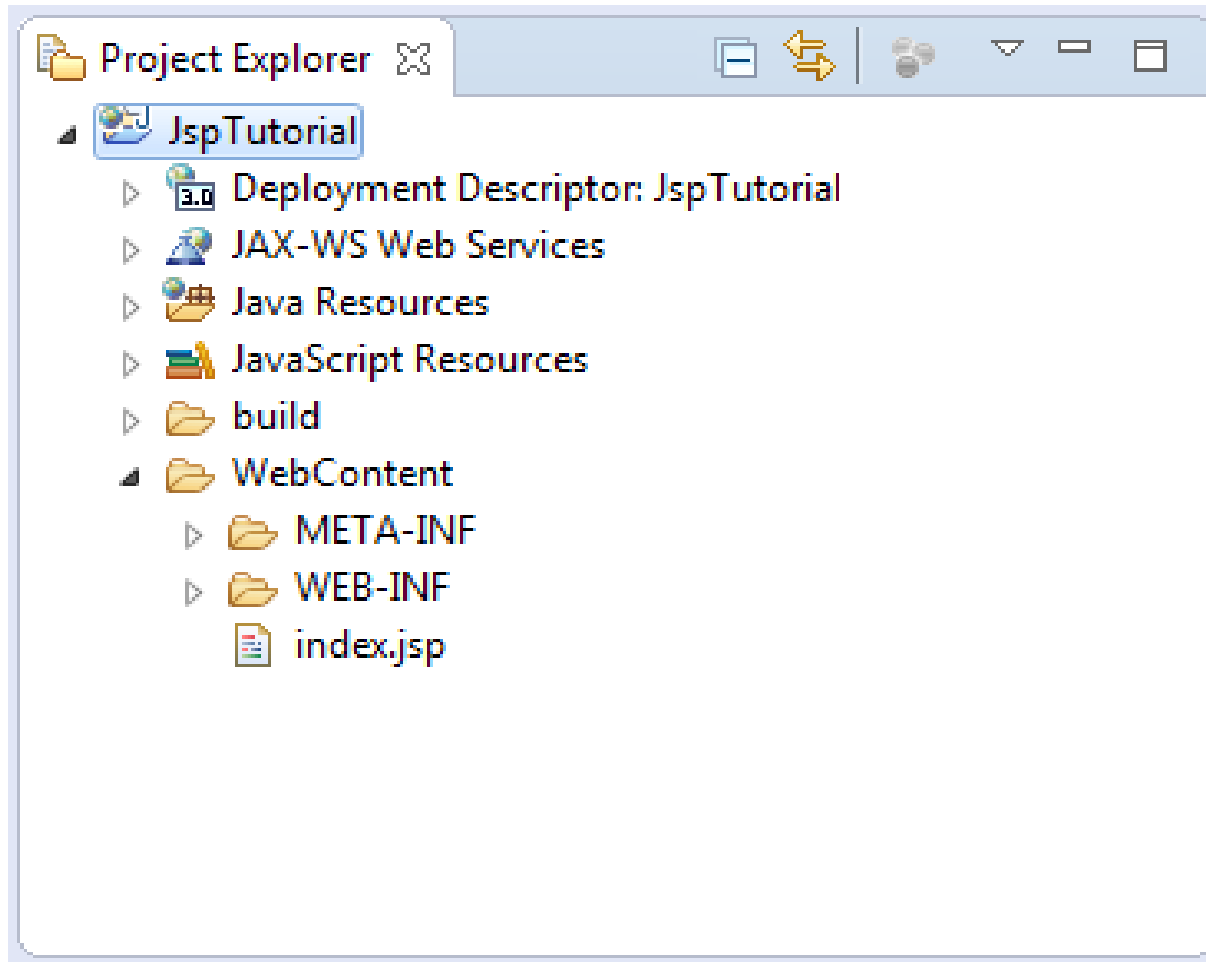


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20.2 Configuring Tomcat Web Server and run WebApp

Now, Project has no error notified.

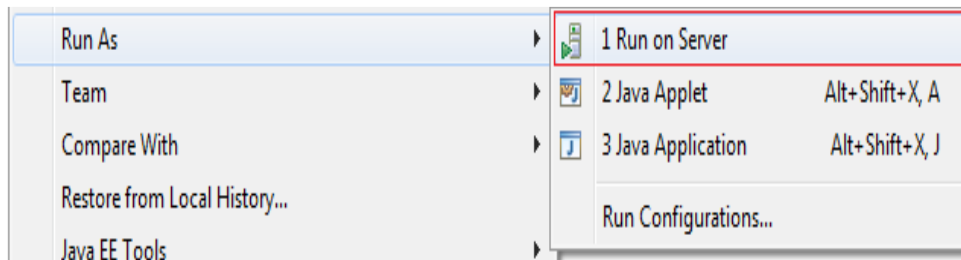


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20.2 Configuring Tomcat Web Server and run WebApp

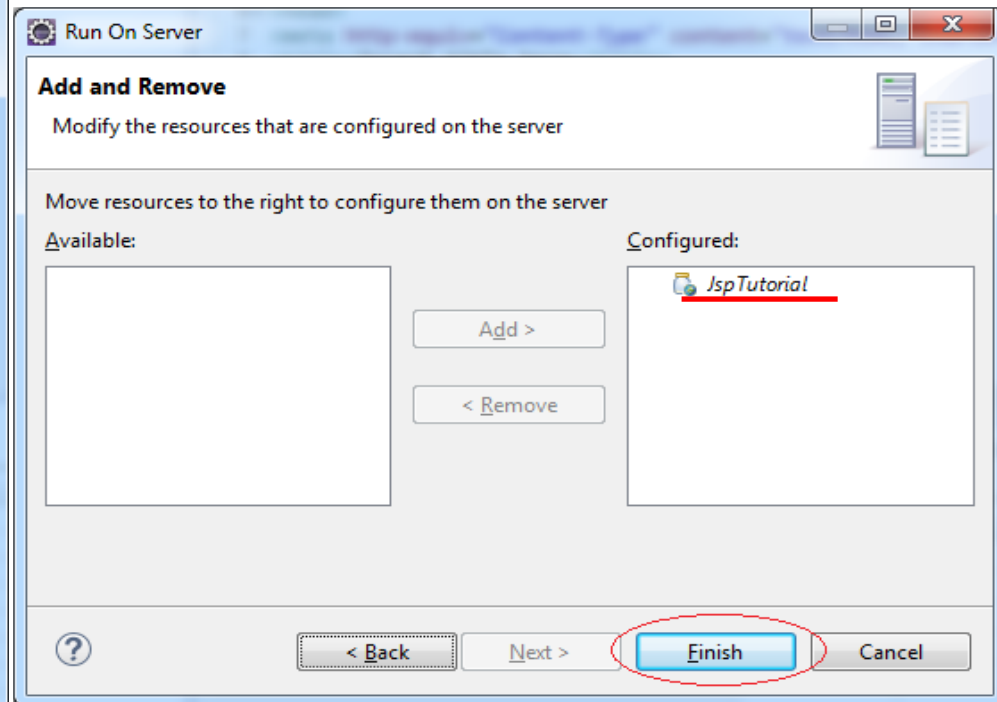
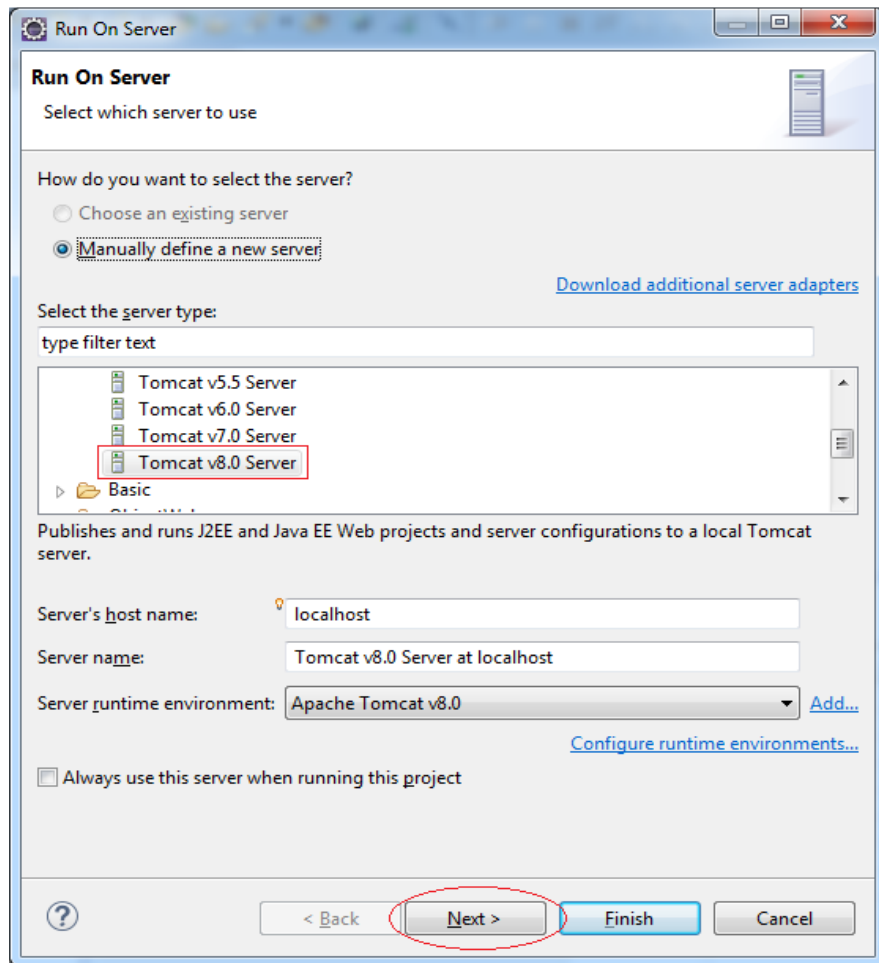
- **Right-click the Project (Jsp Tutorial)**
- and**
- **select Run As/Run on Server.**



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20.2 Configuring Tomcat Web Server and run WebApp

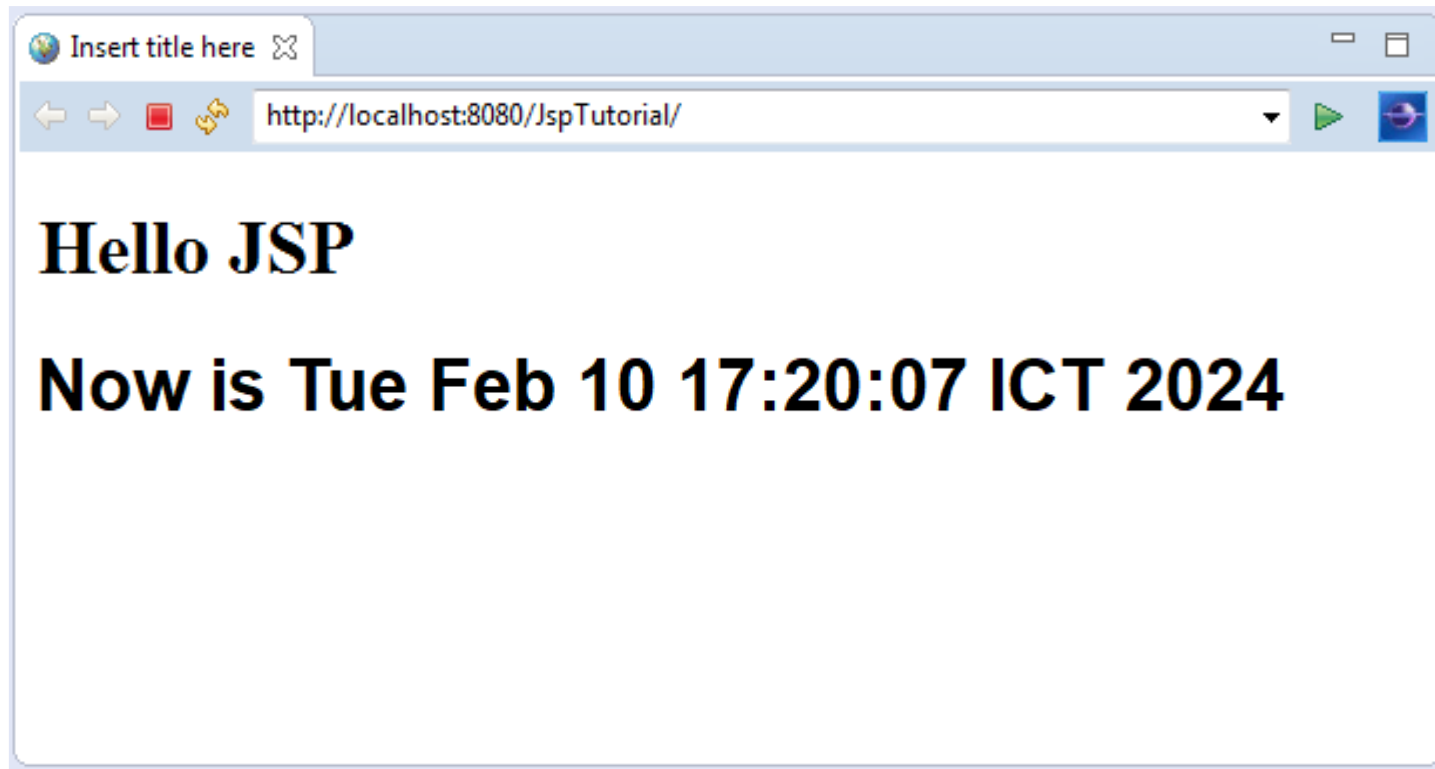


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20.2 Configuring Tomcat Web Server and run WebApp

The results run the application:



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Exercises

1. Bank of Kigali System

Bank of Kigali (BK)
Secure Online Banking

Username
Enter username

Password
Enter password

Login

Demo credentials: Josbert / Josbert

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Bank of Kigali (BK) × +

localhost:8080/Josbert_BK/ Summarize 🔍 ☆ ⋮ Chat

Welcome to Bank of Kigali (BK)

● Dr. N. N. Josbert

Current Balance **FRW 46,500,000.00**

ACCOUNT

Account Holder

Dr. N. N. Josbert

Create Account

DEPOSIT

Amount (FRW)

0.00

Deposit

WITHDRAW

Amount (FRW)

0.00

Withdraw

ACTIONS

Check Balance

Save Transactions (.txt)

Logout

TRANSACTION HISTORY

Account opened	
Deposit	FRW 1,000,000.00
Withdraw	FRW 500,000.00
Deposit	FRW 46,000,000.00

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Bank of Kigali (BK)

localhost:8080/Josbert_BK/

Welcome to Bank of Kigali (BK)

Dr. N. N. Josbert

Current Balance: FRW 46,500,000.00

ACCOUNT
Account Holder: Dr. N. N. Josbert
Create Account

DEPOSIT
Amount (FRW): 0.00
Deposit

WITHDRAW
Amount (FRW): 0.00
Withdraw

ACTIONS
Check Balance
Save Transactions (.txt)
Logout

TRANSACTION HISTORY

Account opened	
Deposit	FRW 1,000,000.00
Withdraw	FRW 500,000.00
Deposit	FRW 46,000,000.00

File Edit View

Bank of Kigali (BK) - Transaction Statement

Account Holder: Dr. N. N. Josbert
Current Balance: FRW 46,500,000.00

Transactions:

Account opened
Deposit FRW 1,000,000.00
Withdraw FRW 500,000.00
Deposit FRW 46,000,000.00

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2. RITCO Ticket Booking System

RITCO Company Ticket Booking

localhost:8080/Josbert_Ritco/

Summarize

Chat

RITCO COMPANY BOOKING TICKET

BOOKED TICKETS

Ticket Code	Name	Phone	Price (RWF)	Status	Route	Date	Time
-------------	------	-------	-------------	--------	-------	------	------

Print Ticket (Text)

Available Buses

5

Available Tickets

20

Booked Tickets

0

Remaining Seats: 20

Book a Ticket

Name:

N.N. Josbert

Phone Number:

+250788547400

Price (RWF):

4000

Status:

Paid

Route:

Kigali - Rubavu

Date:

20/04/2026

Time:

07:57 PM

Ticket Code:

TKT-9225

Book Ticket

localhost:8080 says

Ticket booked successfully!
Code: TKT-9225

OK

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RITCO COMPANY

Company Name:	RITCO
Email:	ritco@gmail.com
Tel:	+250 078000000
Route:	Kigali - Rubavu
Date:	2026-04-20
Time:	19:57
Ticket Code:	TKT-9225
Passenger:	N.N. Josbert
Phone:	+250788547400
Price:	4000 RWF
Status:	Paid
Cashier:	Mukamana Aline

THANK YOU FOR BOOKING WITH US!

Close

jsBc X RITCO_TKT-2 Attendance_ MabestLogis HospitalMan Ev

File Edit View H1

```
#####
RITCO COMPANY
BUS TICKET RECEIPT
#####

Company      : RITCO
Email        : ritco@gmail.com
Tel          : +250 078000000
Country      : RWANDA

=====
TICKET DETAILS
=====
Ticket Code  : TKT-9225
Passenger    : N.N. Josbert
Phone        : +250788547400
Route        : Kigali - Rubavu
Date         : 2026-04-20
Time         : 19:57
Price        : 4000 RWF
Status       : Paid
Cashier      : Mukamana Aline

=====

Printed on   : 2026-04-20 20:01:35

THANK YOU FOR BOOKING WITH RITCO!
Have a safe and pleasant journey!
#####
```

UNIT 9. JSP (Java Server Pages)

3. UoK Attendance System

Bank of Kigali (BK)

Attendance Register – UoK

localhost:8080/JosbertAttendance/

Summarize

Chat

ATTENDANCE MANAGEMENT SYSTEM

UoK ATTENDANCE REGISTER

Dept: Computer Science, SWE Option: Advanced Java Programming Lecturer: Dr. NTEZIRIZA NKERABAHIZI Josbert Date: 2026-04-20

STUDENT NO.

DAY (1-15)

e.g. 2111

e.g. 3

✓ Mark Attendance

+ Add Student

✕ Delete Student

⌘ Calculate & Eligibility

🖨 Print Report

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+ Add New Student

STUDENT NO.

e.g. 2555

FIRST NAME

e.g. Alice

SURNAME

e.g. UWERA

Cancel

Add Student

Exam Eligibility Report

✓ ELIGIBLE ($\geq 80\%$)

Jeanne UWIRINGIYIMANA
2222

86.7%

✗ NOT ELIGIBLE ($< 80\%$)

Olivier KARENZI
2111

6.7%

Rachel MUKESHIMANA
2333

0.0%

Jehovanis KIGELI
2444

0.0%

Close

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4/20/26, 8:15 PM

Attendance Register – UoK

NO.	STUDENT NO.	FIRST NAME	SURNAME	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13
1	2111	Olivier	KARENZI	✓1	0	0	0	0	0	0	0	0	0	0	0	0
2	2222	Jeanne	UWIRINGIYIMANA	✓1	✓1	✓1	0	✓1	✓1	✓1	✓1	✓1	✓1	✓1	✓1	✓1
3	2333	Rachel	MUKESHIMANA	0	0	0	0	0	0	0	0	0	0	0	0	0
4	2444	Jehovanis	KIGELI	0	0	0	0	0	0	0	0	0	0	0	0	0